Calculus II - MATH 141701

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CHAPTER 7: METHODS OF INTEGRATION

Contents

- Review of Substitution and Integration by Tables
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- Summary of Integration Techniques
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We must choose u and compute du when doing **substitution**. Here are some different situations which are useful.

Example

Find

$$\int \frac{x^2}{(x^3-2)^5}\,dx.$$

Solution

Let $u = x^3 - 2$ then $du = 3x^2dx$, so by substitution

$$\begin{aligned}\int \frac{x^2}{(x^3-2)^5}\,dx &= \int \frac{du}{3u^5}\\ &= \frac{u^{-4}}{3(-4)} + C\\ &= \frac{-(x^3-2)^{-4}}{12} + C.\end{aligned}$$

Fitting the form of a known integration formula

Example

Find

$$\int \frac{t}{\sqrt{1-t^4}}dt.$$

Solution

The formula of inverse sine appears if we put $a = 1$, $x = t^2$. We have $dx = 2tdt$,

$$\begin{aligned}\int \frac{t}{\sqrt{1-t^4}}dt &= \int \frac{1}{2\sqrt{1-x^2}}dx \\ &= \frac{1}{2}\sin^{-1}x + C \\ &= \frac{1}{2}\sin^{-1}t^2 + C.\end{aligned}$$

Multiplication by 1 to derive an integration formula

Example

Find

$$\int \sec x dx.$$

Solution

Multiply the integrand by $u := \sec x + \tan x$ and notice that $\sec^2 x + \sec x \tan x = u'$. So,

$$\begin{aligned}\int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx &= \int \frac{du}{u} \\ &= \ln|u| + C \\ &= \ln|\sec x + \tan x| + C.\end{aligned}$$

Substitution after an algebraic manipulation

Example

Find

$$\int \frac{1}{1+e^x}dx.$$

Solution

If we put $u = 1 + e^x$ then $du = e^x dx$ so

$$\int \frac{1}{1+e^x}dx = \int \frac{du}{ue^x}$$

It is not an appropriate form. But if we put $u = 1 + e^{-x}$

$$\int \frac{1}{1+e^x}dx = \int \frac{e^{-x}}{1+e^{-x}}dx = \int \frac{-du}{u}$$
$$= -\ln(e^{-x}+1) + C.$$

Fractional exponents

The integrand involves terms with fractional exponents, we often choose the form $x = u^n$ where n is the least common multiple of the denominators. For example, if the exponential denominators are $4, 3$ and 6, we propose $n = (4, 3, 6) = 12$ and $x = u^{12}$.

Example

Find

$$\int \frac{dx}{x^{1/2} + x^{1/3}}.$$

Solution

If we put $u = x^{1/6}$ then $dx = 6u^5 du$ so

$$\int \frac{dx}{x^{1/2} + x^{1/3}} = \int \frac{6u^5 du}{u^2 + u^3} = \int \frac{6u^3 du}{1 + u}$$

$$\begin{aligned}\int \frac{6u^3 du}{1+u} &= \int \left(6u^2 - 6u + 6 + \frac{-6}{1+u}\right) du \\ &= 2u^3 - 3u^2 + 6u - 6\ln|1+u| + C \\ &= 2x^{1/2} - 3x^{1/3} + 6x^{1/6} - 6\ln(1+x^{1/6}) + C.\end{aligned}$$

Using Tables of Integrals

- Deal with many forms of integrals.
- Using integration tables in **Appendix D**.
- Including **Elemetary (Formulas** $1-29$**), Linear and quadratic (**$30-76$**), Radical (**$77-121$**), Trigonometric (**$122-167$**), Inverse Trigonometric (**$168-182$**), Exponetial and logarithmic (**$183-200$**).**

Example

Find

$$\int (\ln x)^4 dx.$$

Solution: We use formulas 198 **and** 197

$$\begin{aligned}
\int (\ln x)^4 dx &= x(\ln x)^4 - 4\int (\ln x)^3 dx \\
&= x(\ln x)^4 - 4\left[x(\ln x)^3 - 3\int (\ln x)^2 dx\right] \\
&= x(\ln x)^4 - 4x(\ln x)^3 + 12\int (\ln x)^2 dx \\
&= x(\ln x)^4 - 4x(\ln x)^3 + 12\left[x(\ln x)^2 - 2x\ln x + 2x\right] + C \\
&= x(\ln x)^4 - 4x(\ln x)^3 + 12x(\ln x)^2 - 24x\ln x + 24x + C.
\end{aligned}$$

Using an integral table after substitution

Example

Find

$$\int \frac{x}{\sqrt{8-5x^2}}dx.$$

Solution

Let $u = \sqrt{5}x$ then $du = \sqrt{5}dx$. So,

$$\begin{aligned}\int \frac{x}{\sqrt{8-5x^2}}dx &= \int \frac{udu}{5\sqrt{8-u^2}} \\ &= -1/5\sqrt{8-u^2} + C \\ &= -1/5\sqrt{8-5x^2} + C\end{aligned}$$

by Formula 111.

Multiple forms of integrals

Example

Find

$$\int 5x^2\sqrt{3x^2+1}dx.$$

Solution

Let $u = \sqrt{3}x$ then $du = \sqrt{3}dx$. Using formula 87,

$$\begin{aligned}\int 5x^2\sqrt{3x^2+1}dx &= \int \frac{5}{3\sqrt{3}}u^2\sqrt{u^2+1}du \\ &= \frac{5}{3\sqrt{3}}\left[\frac{u(u^2+1^2)^{3/2}}{4} - \frac{u\sqrt{u^2+1}}{8}\right] \\ &- \frac{5}{3\sqrt{3}}\frac{1^4}{8}\ln(u+\sqrt{u^2+1}) + C\end{aligned}$$

$$= \frac{5}{24\sqrt{3}}\left[2\sqrt{3}x(3x^2+1)^{3/2} - \sqrt{3}x\sqrt{3x^2+1}\right]$$
$$-\frac{5}{24\sqrt{3}}\ln(\sqrt{3}x + \sqrt{3x^2+1}) + C.$$

Exercises: 7.1....

Integration by Parts Formula

$$\int u dv = uv - \int v du.$$

Example

Find $\int xe^x dx$.
Solution Let $u = x$, $dv = e^x dx$ then $du = dx$, $v = e^x$, thus

$$\begin{aligned}\int xe^x dx &= xe^x - \int e^x dx \\ &= xe^x - e^x + C.\end{aligned}$$

You might have chosen $u = e^x$, $dv = xdx$
then $du = e^x dx$, $v = \int xdx = \frac{x^2}{2}$,
$\int xe^x dx = e^x \frac{x^2}{2} - \int \frac{x^2}{2} e^x dx$.

Example

Find

$$\int lnxdx\ ,\ x > 0$$

Solution

Let $u = lnx$, $dv = dx$
$du = \frac{1}{x}dx$, $v = x$,

$$\begin{aligned}\int lnxdx &= (lnx)x - \int x(\frac{1}{x})dx \\ &= xlnx - x + C.\end{aligned}$$

Repeated Use of Integration by Parts

Example

Find

$$\int x^2 e^{-x} dx.$$

Solution

Let $u = x^2$, $dv = e^{-x}dx$
then $du = 2xdx$, $v = \int e^{-x}dx = -e^{-x}$ so,

$$\begin{aligned}\int x^2 e^{-x} dx &= x^2(-e^{-x}) - \int (-e^{-x})2xdx \\ &= -x^2 e^{-x} + 2\int xe^{-x}dx \\ &= -x^2 e^{-x} + 2\,x(-e^{-x}) - \int (-e^{-x})dx \\ &= -e^{-x}(x^2 + 2x + 2) + C.\end{aligned}$$

Example

Find

$$\int e^{2x} sinxdx.$$

Solution

Let $u = e^{2x}$, $dv = sinxdx$
then $du = 2e^{2x}dx$, $v = -cosx$ so,

$$\begin{aligned} I = \int e^{2x} sinxdx &= e^{2x}(-cosx) - \int (-cosx)(2e^{2x})dx \\ &= -e^{2x}cosx + 2\int e^{2x}cosxdx \\ &= -e^{2x}cosx + 2\left(e^{2x}(sinx) - \int sinx(2e^{2x})dx\right. \\ &= -e^{2x}cosx + 2e^{2x}sinx - 4\int e^{2x}sinxdx. \end{aligned}$$

Notice $\int e^{2x} sinx dx = I$

$$I = -e^{2x} cosx + 2e^{2x} sinx - 4I$$

$$5I = -e^{2x} cosx + 2e^{2x} sinx + C$$

$$I = \frac{1}{5} e^{2x} (2sinx - cosx) + C$$

Definite Integration by Parts

$$\int_a^b u dv = uv|_a^b - \int_a^b v du.$$

Example

Evaluate

$$\int_0^1 xe^{2x} dx.$$

Solution

Let $u = x$, $dv = e^{2x} dx$

then $du = dx$, $v = \frac{1}{2} e^{2x}$

$$\begin{aligned}\int_0^1 xe^{2x}dx &= \frac{1}{2}xe^{2x}|_0^1 - \frac{1}{2}\int_0^1 e^{2x}dx \\ &= [\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}]_0^1 \\ &= \frac{1}{4}e^2 + \frac{1}{4}.\end{aligned}$$

Example

Evaluate

$$\int_0^1 tan^{-1}x dx.$$

Solution

Let $u = tan^{-1}x$, $dv = dx$

then $du = \dfrac{dx}{1+x^2}$, $v = x$

$$\begin{aligned}\int_0^1 tan^{-1}xdx &= (tan^{-1}x)x|_0^1 - \int_0^1 \frac{xdx}{1+x^2} \\ &= [xtan^{-1}x - \frac{1}{2}ln(1+x^2)]_0^1 \\ &= [1(tan^{-1}1) - \frac{1}{2}ln(1+1)] - [0 - \frac{1}{2}ln1] \\ &= \frac{\pi}{4} - \frac{1}{2}ln2.\end{aligned}$$

Exercises: 7.2....

Powers of Sine and Cosine

- A product of powers of sine and cosine

$$\int \sin^m x \cos^n x dx.$$

- Two cases
  Case I: Either m or n is ood (or both)
  General strategy: Suppose that m is ood. Separate a factor of $\sin x$ from the rest of the integrand, so the remaining power of $\sin x$ is even; use the identity $\sin^2 x = 1 - \cos^2 x$ to express everything but the term $(\sin x dx)$ in terms of $\cos x$.

Example

Find

$$\int \sin^4 x \cos^3 x dx.$$

Solution Since $n = 3$ is odd, use $\cos^2 x = 1 - \sin^2 x$ to express the integral as a polynomial in $\sin x$.

$$\begin{aligned} I = \int \sin^4 x \cos^3 x dx &= \int \sin^4 x \cos^2 x (\cos x dx) \\ &= \int \sin^4 x (1 - \sin^2 x)(\cos x dx) \\ &= \int u^4 (1 - u^2) du \end{aligned}$$

$$\begin{aligned} I &= \frac{1}{5}u^5 - \frac{1}{7}u^7 + C \\ &= \frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + C. \end{aligned}$$

Case II: Both m and n are even

General strategy: Convert this into Case I by using identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x); \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

These are sometimes called half-angle identities.

Example

Find

$$I = \int \sin^2 x \cos^4 x dx.$$

Solution

$$\begin{aligned} I &= \int \sin^2 x \cos^4 x dx = \int \frac{1}{2}(1 - \cos 2x)(\frac{1}{4})(1 + \cos 2x)^2 dx \\ &= \frac{1}{8}\int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx \\ &= \frac{1}{8}\int [1 + \cos 2x - \frac{1}{2}(1 + \cos 4x) - (1 - \sin^2 2x)(\cos 2x)] dx \end{aligned}$$

$$
\begin{aligned}
I &= \frac{1}{8}\int [1 + \cos 2x - \frac{1}{2} - \frac{1}{2}\cos 4x - \cos 2x]\, dx \\
&+ (\sin^2 2x)\cos 2x dx \\
&= \frac{1}{8}\int (\frac{1}{2} - \frac{1}{2}\cos 4x) dx \\
&+ \frac{1}{8}\int \sin^2 2x(\cos 2x) dx \\
&= \frac{1}{16}x - \frac{1}{64}\sin 4x + \frac{1}{48}\sin^3 2x + C.
\end{aligned}
$$

Powers of Secant and Tangent

The simplest integrals of this form are

$$\int tanxdx = ln|secx| + C$$

and

$$\int secdx = ln|secx + tanx| + C.$$

For the more general situation, which we write as

$$\int tan^m xsec^n xdx$$

There are three essentially different cases to consider.

Case I: n is even

General strategy: Peel off a factor of sec^2x from the integrand and use the identity $sec^2x = tan^2x + 1$ to express the integrand in powers of $tanx$ except for (sec^2xdx); substitute $u = tanx$, $du = sec^2xdx$, and integrate by using the power rule.

Example

Find

$$\int tan^2xsec^4xdx.$$

Solution

$$\begin{aligned}
\int tan^2 x sec^4 x dx &= \int tan^2 x sec^2 x (sec^2 x dx) \\
&= \int tan^2 x (tan^2 x + 1) sec^2 x dx \\
&= \int u^2 (1 + u^2) du \\
&= \frac{1}{5} u^5 + \frac{1}{3} u^3 + C \\
&= \frac{1}{5} tan^5 x - \frac{1}{3} tan^3 x + C.
\end{aligned}$$

Case II: m is odd

General strategy: Peel off a factor of $secxtanx$ from the integrand and use the identity $tan^2x = sec^2x - 1$ to express the integrand in powers of $secx$ except for $(secxtanxdx)$; substitute $u = secx$, $du = secxtanxdx$, and integrate by using the power rule.

Example

Find

$$\int tanxsec^6xdx.$$

Solution

$$\begin{aligned}\int tanxsec^6xdx &= \int sec^5x(secxtanxdx)\\ &= \int u^5du\\ &= \frac{1}{6}u^6 + C\\ &= \frac{1}{6}sec^6x + C.\end{aligned}$$

Case III: m is even and n is odd

General strategy: Use the identity $tan^2x = sec^2x - 1$ to express the integrand in powers of $secx$; then use the reduction Formula 161:

$$\int \sec^n(au)du = \frac{\sec^{n-2}(au)\tan(au)}{a(n-1)} + \frac{n-2}{n-1}\int \sec^{n-2}(au)du$$

Example

Find

$$\int tan^2xsec^3xdx.$$

Solution

$$\int tan^2 x sec^3 x dx = \int (sec^2 x - 1) sec^3 x dx \qquad = [\frac{sec}{}$$

$$= \frac{sec^3 x tanx}{4} - \frac{1}{4}\int sec^3 x dx$$

$$= \frac{sec^3 x tanx}{4} - \frac{1}{4}[\frac{secxtanx}{2} + \frac{1}{2}\int secxdx]$$

$$= \frac{sec^3 x tanx}{4} - \frac{secxtanx}{8} - \frac{1}{8} ln|secx + tanx| + C.$$

Trigonometric substitutions

Trigonometric substitutions can also be useful. For instance, suppose an integrand contains the term $\sqrt{a^2 - u^2}$ where $a > 0$. Then by setting $u = a \sin\theta$ for an acute angle θ, and using the identity $\cos^2\theta = 1 - \sin^2\theta$, we obtain

$$\sqrt{a^2 - u^2} = \sqrt{a^2 - a^2\sin^2\theta} = a\sqrt{1 - \sin^2\theta} = a\cos\theta$$

Thus, the substitution $u = a\sin\theta$, $du = a\cos\theta d\theta$ eliminates the square root and may covert the given integral into one involving only sine and cosine. This substitution can best be remembered by setting up a reference triangle. This process is illustrated in the following example.

Example

Find

$$\int x^3\sqrt{x^2-1}dx.$$

Solution

Let $x = sec\theta$, $dx = sec\theta tan\theta d\theta$; we use the reference triangle

$$\begin{aligned} I = \int x^3\sqrt{x^2-1}dx &= \int sec^3\theta\sqrt{sec^2\theta - 1}(sec\theta tan\theta d\theta) \\ &= \int sec^4\theta tan^2\theta d\theta \\ &= \int sec^2\theta tan^2\theta(sec^2\theta d\theta) \end{aligned}$$

$$
\begin{aligned}
I &= \int (tan^2\theta + 1)tan^2\theta(sec^2\theta d\theta) \\
&= \int (u^4 + u^2)du \\
&= \frac{1}{5}u^5 + \frac{1}{3}u^3 + C \\
&= \frac{1}{5}tan^5\theta + \frac{1}{3}tan^3\theta + C \\
&= \frac{1}{5}(x^2 - 1)^5/2 + \frac{1}{3}(x^2 - 1)^3/2 + C.
\end{aligned}
$$

Quadratic-Form integrals

Example

Find

$$\int \sqrt{16x - 2x^2 - 23}dx.$$

Solution

Complete the square within the radical

$$\begin{aligned} 16x - 2x^2 - 23 &= -2(x^2 - 8x) - 23 \\ &= -2(x^2 - 8x + 4^2) + 2.4^2 - 23 \\ &= -2(x-4)^2 + 9 \end{aligned}$$

$$\int \sqrt{16x - 2x^2 - 23}dx = \int \sqrt{9 - 2(x-4)^2}dx = \int \sqrt{9 - u^2}(\frac{du}{\sqrt{2}})$$

$$= \int \sqrt{9 - (3sin\theta)^2}(\frac{3}{\sqrt{2}}cos\theta d\theta)$$

$$= \frac{3}{\sqrt{2}}\int 3\sqrt{1 - sin^2\theta}cos\theta d\theta$$

$$= \frac{9}{\sqrt{2}}\int cos^2\theta d\theta = \frac{9}{2\sqrt{2}}[\theta + \frac{sin2\theta}{2}] + C$$

$$= \frac{9}{2\sqrt{2}}sin^-1[\frac{\sqrt{2}}{3}(x-4)]$$

$$+ \frac{x-4}{2}\sqrt{16x - 2x^2 - 23} + C.$$

Partial Fractional Decomposition

- We only deal with the reduced fraction since polynomials are easy to integrate.
- A single linear power in Denominator

$$f(x) = \frac{P(x)}{(x-r)^n}.$$

- A single quadratic factor a in Denominator

$$f(x) = \frac{P(x)}{(x^2+sx+t)^n}.$$

- Combine two above forms.

Linear Factor

Let $f(x) = \frac{P(x)}{(x-r)^n}$, where $d(P) < n$, $P(r) \neq 0$ then f can be decomposed into fractions in the following form

$$f(x) = \frac{A_1}{(x-r)} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_n}{(x-r)^n}$$

where $A_1, A_2, \cdots, A_n$ are constant.

Example

Decompose $\frac{x^2-6x+3}{(x-2)^3}$.

Solution

$$\frac{x^2-6x+3}{(x-2)^3} = \frac{A_1}{(x-2)} + \frac{A_2}{(x-2)^2} + \frac{A_3}{(x-2)^3}$$
$$x^2-6x+3 = A_1(x-2)^2 + A_2(x-2) + A_3.$$

Let $x = 2$ imply $A_3 = -5$. Then we have

$$x^2-6x+3 = A_1x^2 - (4A_1 - A_2)x + 4A_1 - 2A_2 - 5$$

yields $A_1 = 1$, $4A_1 - A_2 = 6$, $4A_1 - 2A_2 - 5 = 3$, thus

$$A_1 = 1,\ A_2 = -2,\ A_3 = -5.$$

If the denominator is the product of n distinct linear factors then we could decompose it into the form

$$f(x) = \frac{P(x)}{(x-r_1)\cdots(x-r_n)} = \frac{A_1}{x-r_1} + \cdots + \frac{A_n}{x-r_n}.$$

Example

Decompose $\frac{8x-1}{x^2-x-2}$.

Solution

$$\frac{8x-1}{x^2-x-2} = \frac{a}{x-2} + \frac{b}{x+1} = \frac{(a+b)x+a-2b}{x^2-x-2}$$

implies $a+b=8$, $a-2b=-1$. Thus, $a=5$, $b=3$

$$\frac{8x-1}{x^2-x-2} = \frac{5}{x-2} + \frac{3}{x+1}.$$

If the denominator consists of mixture of distinct and repeated linear factor, we combine the procedures in previous examples. For example, to decompose

$$\frac{P(x)}{(x-r)^n(x-b)} = \frac{A_1}{(x-r)} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_n}{(x-r)^n} + \frac{A_{n+1}}{x-b}.$$

Note that if the degree of denominator is N then we need to find N constant numbers.

Quadratic Factor

Let $f(x) = \frac{P(x)}{(x^2+sx+t)^n}$, where $d(P) < 2n$ then f can be decomposed into fractions in the following form

$$f(x) = \frac{A_1x + B_1}{x^2 + sx + t} + \frac{A_2x + B_2}{(x^2 + sx + t)^2} + \cdots + \frac{A_nx + B_n}{(x^2 + sx + t)^n}$$

where $A_1, A_2, \cdots, A_n,\ B_1, B_2, \cdots, B_n$ are constant.

Example

Decompose $\frac{-3x^3-x}{(x^2+1)^2}$.

Solution

$$\begin{aligned}\frac{-3x^3 - x}{(x^2+1)^2} &= \frac{A_1x + B_1}{x^2+1} + \frac{A_2x + B_2}{(x^2+1)^2} \\ &= \frac{(A_1x + B_1)(x^2+1) + (A_2x + B_2)}{(x^2+1)^2} \\ &= \frac{A_1x^3 + B_1x^2 + (A_1 + A_2)x + B_1 + B_2}{(x^2+1)^2}\end{aligned}$$

imply

$$A_1 = -3,\ B_1 = 0,\ A_2 = 2,\ B_2 = 0.$$

SUMMARY

Let $f(x) = \frac{P(x)}{Q(x)}$, where $d(P) < d(Q)$ and Q can be written in form of product of linear and irreducible quadratic powers. Then we can decompose $f(x)$ into the sum of forms

$$\frac{A_i}{(x-r)^i}, \quad \frac{A_m x + B_m}{(x^2 + sx + t)^m}.$$

Integrating Rational Functions

Example

$$\int \frac{x^2-6x+3}{(x-2)^3}dx = \int \left[\frac{1}{(x-2)} + \frac{-2}{(x-2)^2} + \frac{-5}{(x-2)^3}\right] dx$$
$$= \ln|x-2| + \frac{2}{x-2} + \frac{5}{2(x-2)^2} + C.$$

$$\int \frac{-3x^3-x}{(x^2+1)^2}dx = \int \left[\frac{-3x}{x^2+1} + \frac{2x}{(x^2+1)^2}\right] dx$$
$$= \int u^{-2} - 3/2u^{-1}du \quad (u = x^2+1, du = 2xdx)$$
$$= \frac{-1}{x^2+1} - \frac{3}{2}\ln(x^2+1) + C.$$

Example

Find

$$I = \int \frac{x^2 + 4x - 23}{(x^2+4)(x+3)}.$$

Solution Decompose the integrand

$$\frac{x^2 + 4x - 23}{(x^2+4)(x+3)} = \frac{3x-5}{x^2+4} + \frac{-2}{x+3}.$$

We have

$$\begin{aligned} I &= 3\int \frac{x}{x^2+4}dx - 5\frac{dx}{x^2+4} - 2\int \frac{dx}{x+3} \\ &= \frac{3}{2}\ln(x^2+4) - \frac{5}{2}\sin^{-1}\frac{x}{2} - 2\ln|x+3| + C. \end{aligned}$$

Rational Functions of sine and cosine

WEIERSTRASS SUBSTITUTION

For $-\pi < x < \pi$, let $u = \tan\frac{x}{2}$ so that

$$\sin x = \frac{2u}{1+u^2},\ \cos x = \frac{1-u^2}{1+u^2},\ dx = \frac{2}{1+u^2}du.$$

This is called the **Weierstrass substitution**.

Example

Find

$$W = \int \frac{dx}{3\cos x - 4\sin x}.$$

Solution

Applying Weierstrass substitution,

$$
\begin{aligned}
W &= \int \frac{\frac{2du}{1+u^2}}{3\frac{1-u^2}{1+u^2} - 4\frac{2u}{1+u^2}} \\
&= \int \frac{-2du}{3u^2+8u-3} = \int \frac{-2du}{(3u-1)(u+3)} \\
&= \int \left(\frac{-3}{5(3u-1)} + \frac{1}{5(u+3)} \right) du \\
&= -\frac{1}{5} \ln|3u-1| + \frac{1}{5} \ln|u+3| + C \\
&= -\frac{1}{5} \ln\left|3\tan\frac{x}{2} - 1\right| + \frac{1}{5} \ln\left|\tan\frac{x}{2} + 3\right| + C.
\end{aligned}
$$

Integration strategy

- Step 1: Simplify
  Simplify the integrand, if possible, and use one of the procedural rules.
- Step 2: Use basic formulas
  Use the basic integration formulas.
- Step 3: Substitute.
  Make any substitution that will transform the integral into one of the basic form.
- Step 4: Classify

I. Integration by parts

- A. Form $\int x^n e^{ax} dx, \int x^n sinaxdx, \int x^n cosaxdx$
  Let $u = x^n$
- B. Form $\int x^n lnxdx, \int x^n sin^-1axdx, \int x^n tan^-1axdx$
  Let $dv = x^n dx$
- C. Form $\int sinbxe^{ax} dx, \int e^{ax} cosbxdx$
  Let $dv = e^{ax} dx$

II. Trigonometric forms

A. $\int sin^m x cos^n x dx$

- m odd:
  Peel off a factor of $(sinxdx)$ and let $u = cosx$.
- n odd
  Peel off a factor of $(cosxdx)$ and let $u = sinx$.
- n and m both even:
  Use half-angle identities:
  $cos^2 x = \frac{1}{2}(1 + cos2x)$ and $sin^2 x = \frac{1}{2}(1 - cos2x)$.

B. $\int tan^m x sec^n x dx$

- n even:
  Peel off a factor of $(sec^2 dx)$ and let $u = tanx$.
- m odd
  Peel off a factor of $(secxtanxdx)$ and let $u = secx$.
- m even, n odd:
  Write using powers of secant and use a reduction formula.

C. For a rational trigonometric integral, try the Weierstrass substituon. Let $u = tan\frac{x}{2}$ so that: $sinx = \frac{2u}{1+u^2}$; $cosx = \frac{1-u^2}{1+u^2}$; $dx = \frac{2}{1+u^2}du$

III. Radical form; try a trigonometric substitution.

A. Form $\sqrt{a^2 - u^2}$: Let $u = asin\theta$

B. Form $\sqrt{a^2 + u^2}$: Let $u = atan\theta$

C. Form $\sqrt{u^2 - a^2}$: Let $u = asec\theta$

IV. Rational forms; try partial fraction decomposition.
Step 5 Try again. Still stuck?

- 1. Manipulate the integrand:
  Multiply by 1
  Rationalize the numerator
  Rationalize the denominator.
- 2. Relate the problem to a previously worked problem
- 3. Look at another table of integrals or consult computer software that does integration.
- 4. Some intergrals do not have simple antiderivatives. We will look at some of these forms later in the text
- 5. If dealing with a definite integral, an approximation may suffice.

- First-Order **Linear** Differential Equations.
- Applications.

- The general form of The First-Order Linear Differential Equation

$$\frac{dy}{dx} + P(x)y = Q(x).$$

Theorem

The general solution of the first-order linear differential equation is

$$y(x) = \frac{1}{I(x)}\left(\int I(x)Q(x)dx + C\right)$$

where $I(x) = e^{\int P(x)dx}$ *is called an "integrating factor" and* C *is an arbitrary constant.*

Proof: Direct calculation.

$1^{st} - order$ initial value problem: first-order differential equation and value of function y at $x = x_0$. We follow such problems by considering some examples.

Example

Solve $y'(x) = e^{-x} - 2y, \ x \geq 0$ and $y(0) = 2$.

Solution

Rewrite equation of the form $y' + 2y = e^{-x}$.

So, we have $P(x) = 2$ and $Q(x) = e^{-x}$.

The integrating factor

$$I(x) = e^{\int 2dx} = e^{2x}.$$

General solution of our equation is

$$y(x) = e^{-2x}\left(\int e^{2x}e^{-x} + C\right) = e^{-2x}(e^{x} + C).$$

Note that $y(0) = 2$ imply $C = 1$. Thus,

$$y = e^{-x} + e^{-2x}, \qquad x \geq 0.$$

Example

Solve the following differential equation

$$x\frac{dy}{dx} + y = xe^{2x}, \quad x > 0.$$

Solution

- Rewrite equation $y' + y/x = e^{2x}$, $P(x) = 1/x$, $Q(x) = e^{2x}$.
- The integrating factor is

$$I(x) = e^{\int P(x)dx} = e^{\int 1/x dx} = e^{\ln|x|} = x.$$

- The general solution is

$$y = \frac{1}{x}\left(\int xe^{2x}dx + C\right) = \frac{1}{x}\left(\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C\right)$$

by integrating by parts.

Applications

- Modelling Logistic Growth.
- Modelling Dilution.
- Modelling RL Circuits.

Logistic Growth

- Denote $Q(t)$ be the population of a colony of living organisms. When $Q(t)$ is small, the relative rate of change of population is constant. So,
$$\frac{dQ}{dt} = kQ$$
this is called **exponential growth**.
- If the population has **carrying capacity** B of the species. We assume the relative rate of change of population is proportional to the difference $B - Q$. Thus that leads to equation
$$\frac{dQ}{dt} = kQ(B - Q)$$
this is called **logistic equation**.

Example

The rate at which an epidemic spreads through a community is proportional to the product of the number of residents who have been infected and the number of sesceptible residents. Express the number of residents who have bee infected as a function of time.

Solution

$Q(t)$: infected residents;

B : total residents;

$B - Q(t)$: susceptible residents;

The differential equation

$$\frac{dQ}{dt} = kQ(B - Q).$$

Solving the above equation by using separation of variables

$$Q(t) = \frac{B}{1 + Ae^{-Bkt}}$$

where A is an arbitrary constant.

Modelling Dilution

Example

A tank contains 20 lb of salt dissolved in 50 gal of water. Suppose 3 gal of brine containing 2 lb of dissolved salt per gallon runs into the tank every minute and that the mixture (kept uniform by stirring) runs out of tank at rate of 2 gal/min. Find the amount of salt in the tank at any time t. How much salt is in the tank at the end of one hour?

Solution

$S(t)$ the amount of salt in the tank. Each minute 3 gal of brine flows into the tank and each gallon contains 2 lb of salt, thus, the inflow rate is 6 lb/min.

At time t, there are $50 + (3 - 2)t$ gallons of water in the tank and $S(t)$ lb of salt, yields the concentration of salt in the water at time t is

$$\frac{S(t)}{50 + t} \quad lb/gal$$

The outflow rate of salt is
$\left(\frac{S(t)}{50+t}\ lb/gal\right)(2\ gal/min) = \frac{2S(t)}{50+t} \quad lb/min.$
The net rate of change of salt is dS/dt or

$$\frac{dS}{dt} = 6 - \frac{2S}{50 + t}$$

this is the first-order linear differential equation with
$P(t) = 2/(50 + t),\ Q(t) = 6.$

Solving the above equation

$$S(t) = 2(50+t) + \frac{C}{(50+t)^2}.$$

Since $S(0) = 20$, then $C = -200000$ and solution

$$S(t) = 2(50+t) - \frac{200000}{(50+t)^2}.$$

At the end of one hour, the tank contains

$$S(60) = 2(50+60) - \frac{200000}{(50+60)^2} \approx 203.471 \ lb.$$

Modelling RL Circuits

- The first-order differential equation

$$L\frac{dI}{dt} + RI = E.$$

where R resistance, L inductance, $I(t)$ current at time t.

- Solution

$$I(t) = \frac{E}{R} + Ce^{-(R/L)t}.$$

- At $t = 0$, $I(0) = 0$ then $C = -E/R$. Thus,

$$I = \frac{E}{R}\left(1 - e^{-Rt/L}\right).$$

- As time tends to infinity, the current approaches E/R **steady-state current**.

IMPROPER INTEGRALS

- Type I (Infinite limits of integration).
- Type II (Unbounded integrands).
- Comparison test.

Type I

Definition

(a) If $\int_a^N f(x)\,dx$ exists $(N \geq a)$ then

$$\int_a^{\infty} f(x)\,dx := \lim_{N\to\infty} \int_a^N f(x)\,dx$$

(b) If $\int_t^b f(x)\,dx$ exists $(t \leq b)$ then

$$\int_{-\infty}^{b} f(x)\,dx := \lim_{t\to-\infty} \int_t^b f(x)\,dx$$

$\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^{b} f(x)\,dx$ are called CONVERGENT if the corresponding limit exists (a finite number) and DIVERGENT if the limit does not exist.

Type I

Definition

(c) If both $\int_a^\infty f(x)\,dx$ and $\int_{-\infty}^a f(x)\,dx$ are convergent then we define

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_a^{\infty} f(x)\,dx + \int_{-\infty}^{a} f(x)\,dx$$

- Any real number a can be used in (c).

Example

- Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$.
- **Solution:** Part (a) of Definition, we have

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{N\to\infty}\int_1^N \frac{1}{x^2}\,dx = \lim_{N\to\infty} \frac{-1}{x}\Big|_1^N = 1$$

$\Longrightarrow$ CONVERGENT !!!

Example

- $\int_1^\infty \frac{1}{x}\,dx$ is convergent or divergent ?
- **Solution:** Part (a) of Definition, we have

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} \ln|x|\Big|_1^t = \infty$$

$\Longrightarrow$ DIVERGENT !!!

Example

- Find all values of p such that $\int_1^\infty \frac{1}{x^p}\,dx$ is convergent.
- $$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{N\to\infty} \frac{1}{1-p}\left[\frac{1}{N^{p-1}} - 1\right]$$
- $\int_1^\infty \frac{1}{x^p}\,dx$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Example

- Compute $\int_0^\infty xe^{-2x}\,dx$ if convergence.
- Integrate by parts
  $u = x,\ dv = e^{-2x}dx \Rightarrow du = dx,\ v = -1/2e^{-2x}$

$$\int_0^N xe^{-2x}\,dx = -1/2xe^{-2x}\,|_0^N + \int_0^N 1/2e^{-2x}\,dx$$

$$= -\frac{N}{2e^{2N}} - \frac{1}{4e^{2N}} + \frac{1}{4}.$$

- The first term tends to 0 by L'Hospital rule.
- $$\int_0^\infty xe^{-2x}\,dx = \frac{1}{4}.$$

Example

Gabriel's horn is the solid formed by revolving about the x-axis the unbounded region under the curve $y = 1/x$ for $x \geq 1$. Show that this solid has finite volume but infinite surface area.

Solution

The volume

$$V = \int_1^{\infty} \pi x^{-2} dx = \pi \lim_{N \to \infty} \int_1^{N} x^{-2} dx = \pi.$$

The surface area

$$A = 2\pi \int_1^{\infty} f(x)\sqrt{1 + f'(x)^2} = 2\pi \int_1^{\infty} \frac{1}{x}\sqrt{\frac{x^4+1}{x^4}}.$$

Substitution $u = x^2$ then

$$\begin{aligned} A &= 2\pi \int_1^{\infty} \frac{\sqrt{u^2+1}}{2u^2} du \\ &= \pi \lim_{N\to\infty} \int_1^{N} \frac{\sqrt{u^2+1}}{u^2} du \\ &= \pi \lim_{N\to\infty} \left[\frac{-\sqrt{u^2+1}}{u} + \ln(u + \sqrt{u^2+1}) \right]_1^N \\ &= \infty. \end{aligned}$$

Example

- Compute $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$ if convergence.
- $\int_{-\infty}^{\infty} xe^{-x^2}\,dx = \int_{-\infty}^{0} xe^{-x^2}\,dx + \int_{0}^{\infty} xe^{-x^2}\,dx = 0.$
- Integration by substitution $u = x^2 \Rightarrow du = 2xdx$
- $$\int_0^{\infty} xe^{-x^2}\,dx = \lim_{t\to\infty}\int_0^t xe^{-x^2}\,dx = \lim_{t\to\infty}\int_0^{t^2} \frac{1}{2}e^{-u}\,du = \frac{1}{2}.$$
- $$\int_{-\infty}^{0} xe^{-x^2}\,dx = \lim_{t\to-\infty}\int_{t^2}^{0} \frac{1}{2}e^{-u}\,du = \frac{-1}{2}.$$

Example

Evaluate

$$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx \quad \text{and} \quad \int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$$

- Since $y = \frac{1}{1+x^2}$ is symmetric about the y-axis then $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = 2\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx$.
- $$\begin{aligned}\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx &= \lim_{t\to-\infty} \int_{t}^{0} \frac{1}{1+x^2}\,dx \\ &= \lim_{t\to-\infty} \left(\tan^{-1} 0 - \tan^{-1} t\right) = \frac{\pi}{2}.\end{aligned}$$

Type II

Definition

(a) If f is continuous on $[a, b)$ and discontinuous at b then

$$\int_a^b f(x)\,dx := \lim_{t \to b^-} \int_a^t f(x)\,dx$$

(b) If f is continuous on $(a, b]$ and discontinuous at a then

$$\int_a^b f(x)\,dx := \lim_{t \to a^+} \int_t^b f(x)\,dx$$

$\int_a^b f(x)\,dx$ is called CONVERGENT if the corresponding limit exists (a finite number) and DIVERGENT if the limit does not exist.

Type II

Definition

(c) If f has a discontinuity at $c \in (a, b)$ and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Example

Evaluate

$$\int_0^1 \frac{dx}{(x-1)^{2/3}}.$$

Solution

$$\begin{aligned}\int_0^1 \frac{dx}{(x-1)^{2/3}} &= \lim_{t\to 1^-} \int_0^t \frac{dx}{(x-1)^{2/3}} \\ &= \lim_{t\to 1^-} \left[3(x-1)^{1/3}\right]_0^t \\ &= 3.\end{aligned}$$

Example

Evaluate

$$\int_{\pi/2}^{\pi} \sec x dx.$$

Solution

$$\begin{aligned}
\int_{\pi/2}^{\pi} \sec x dx &= \lim_{t \to \pi/2^+} \int_t^{\pi} \sec x dx \\
&= \lim_{t \to \pi/2^+} \left[\ln|\sec x + \tan x|\right]_t^{\pi} \\
&= \lim_{t \to \pi/2^+} \left[\ln|-1+0| - \ln|\sec t + \tan t|\right]_t^{\pi} \\
&= -\infty.
\end{aligned}$$

Example

Evaluate

$$\int_0^3 \frac{dx}{x-2}.$$

Solution

$\int_0^3 \frac{dx}{x-2} = \int_0^2 \frac{dx}{x-2} + \int_2^3 \frac{dx}{x-2}.$

$$\begin{aligned}\int_0^2 \frac{dx}{x-2} &= \lim_{t\to 2^-} \int_0^t \frac{dx}{x-2} \\ &= \lim_{t\to 2^-} \left[\ln|t-2|\right]_0^t \\ &= -\infty.\end{aligned}$$

So, the integral is divergent.

Comparison Test

Theorem

f, g are continuous functions and $0 \leq g(x) \leq f(x)$, $x \geq a$. Then

(a) *If $\int_a^\infty f(x)\,dx$ is convergent then $\int_a^\infty g(x)\,dx$ is convergent.*

(b) *If $\int_a^\infty g(x)\,dx$ is divergent then $\int_a^\infty f(x)\,dx$ is divergent.*

Example

Show that

$$\int_1^{\infty} e^{-x^2} dx$$

converges.

Solution

We have $0 < e^{-x^2} < 1, e^{-x}$ for $x > 1$.

$$\begin{aligned}\int_1^{\infty} e^{-x} dx &= \lim_{N\to\infty} \int_1^{N} e^{-x} dx \\ &= \lim_{N\to\infty} \left[-e^{-x}\right]_1^N \\ &= e^{-1}.\end{aligned}$$

By Comparison Theorem, we obtain the desired result.

Hyperbolic functions

Definition

We define

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

$$\coth x = \frac{1}{\tanh x}, \qquad sechx = \frac{1}{\cosh x}, \quad cschx = \frac{1}{\sinh x}.$$

Theorem

Properties of hyperbolic functions

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh(-x) = -\sinh x$$

$$\cosh(-x) = \cosh x$$

$$\tanh(-x) = -\tanh x$$

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

Proof: Fundamental calculations.

Theorem

Rules for differentiating the hyperbolic functions

$$(\sinh u)' = (\cosh u)\frac{du}{dx}, \qquad (\cosh u)' = (\sinh u)\frac{du}{dx}$$

$$(\tanh u)' = (sech^2\, u)\frac{du}{dx}, \qquad (\coth u)' = -(csch^2\, u)\frac{du}{dx}$$

$$(sech\, u)' = -(sech\, u)(\tanh u)\frac{du}{dx}$$

$$(cschu)' = -(csch\, u)(\coth u)\frac{du}{dx}.$$

Proof: Fundamental calculations.

Theorem

Rules for integrating the hyperbolic functions

$$\int \sinh x dx = \cosh u + C, \qquad \int \cosh x dx = \sinh x + C$$

$$\int sech^2 x dx = \tanh x + C, \qquad \int csch^2 x dx = -\coth x + C$$

$$\int (sech\, x)(\tanh x) dx = -sech\, x + C$$

$$\int (csch\, x)(\coth x) dx = -cschx + C.$$

Proof: Because of differentiability.

Inverse Hyperbolic Functions

Theorem

Logarithmic formulas

$$\sinh^{-1} x = \ln(x + \sqrt{x^2+1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2-1}) \ (x \geq 1)$$

$$\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}, \ (|x| < 1)$$

$$\coth^{-1} x = \frac{1}{2} \ln \frac{1+x}{x-1}, \ (|x| > 1)$$

$$sech^{-1} x = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right), \ csch^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right).$$

Theorem

Rules for differentiating the inverse hyperbolic functions

$$(\sinh^{-1} u)' = \frac{1}{\sqrt{1+u^2}}\frac{du}{dx}, \qquad (\cosh^{-1} u)' = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}$$

$$(\tanh^{-1} u)' = \frac{1}{1-u^2}\frac{du}{dx}, \qquad (\coth^{-1} u)' = \frac{1}{1-u^2}\frac{du}{dx}$$

$$(sech^{-1} u)' = \frac{-1}{u\sqrt{1-u^2}}\frac{du}{dx}$$

$$(csch^{-1} u)' = \frac{-1}{|u|\sqrt{1+u^2}}\frac{du}{dx}.$$

Proof: Fundamental calculations.

Theorem

Rules for integration of the inverse hyperbolic functions

$$\int \frac{1}{\sqrt{1+u^2}}du = \sinh^{-1} u + C, \quad \int \frac{1}{\sqrt{u^2-1}}du = \cosh^{-1} u + C$$

$$\int \frac{1}{1-u^2}du = \tanh^{-1} u + C, \quad \int \frac{1}{1-u^2}du = \coth^{-1} u + C$$

$$\int \frac{-1}{u\sqrt{1-u^2}}du = -sech^{-1}|u|) + C$$

$$\int \frac{-1}{|u|\sqrt{1+u^2}}du = -csch^{-1}|u| + C.$$

Proof: Fundamental calculations.